

Unit 4: Systematic Selection

- A. Principles
- B. Fractional Intervals
- C. Implicit Stratification
- D. Variance Estimation

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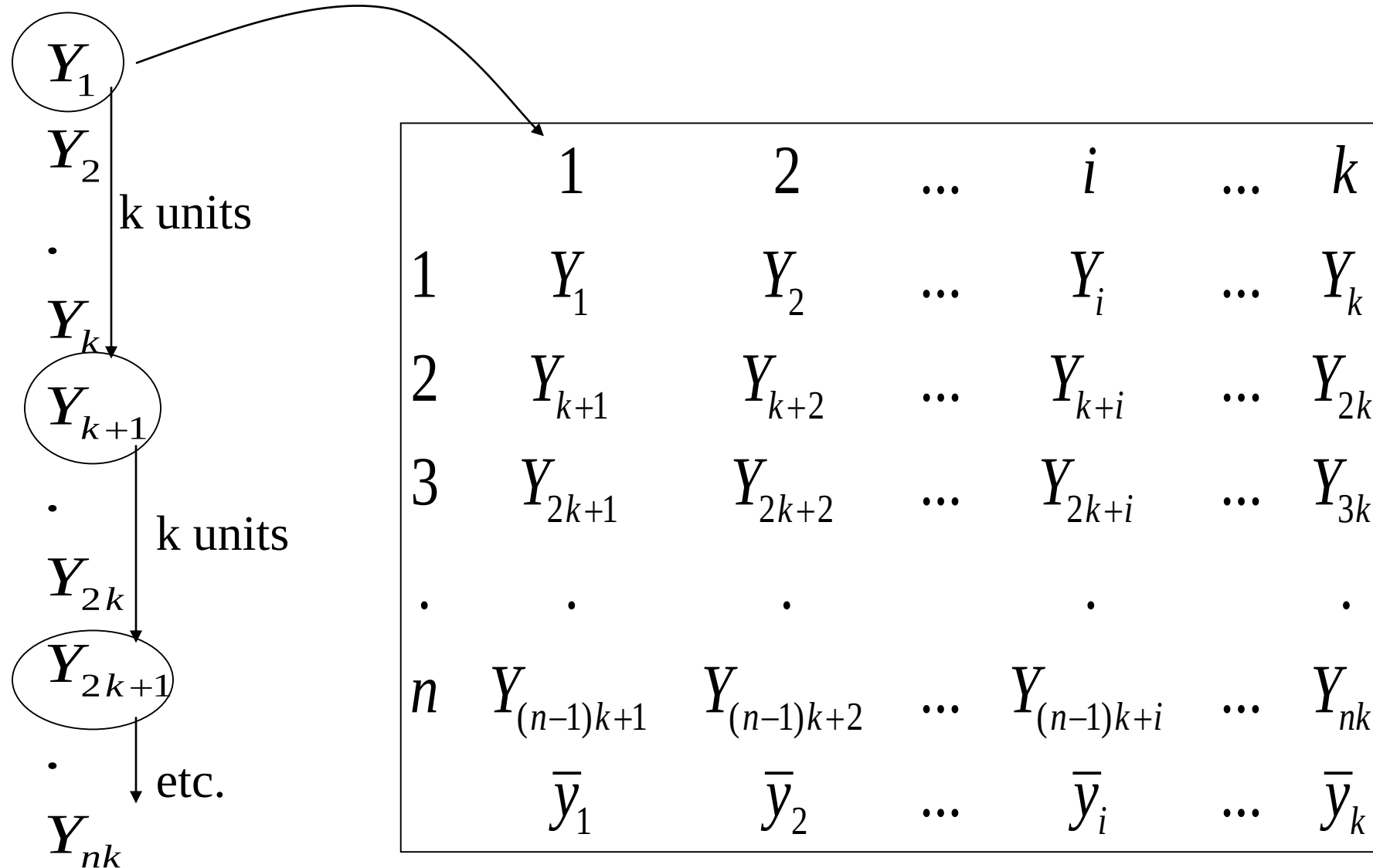
A. Principles

- SRS is difficult to implement and check
 - Without replacement selection
- Systematic sampling is clerically easy to do
 - There are problems in implementation
- Consider systematic sampling from a numbered list with N population elements, n sample elements, and $f = n/N$

Sample selection

- The sampling interval is $I = \frac{N}{n} = \frac{1}{f} = F = k$
- Selection starts with choosing a random start RS from 1 to k
 - Select population elements RS, RS+ k , RS+2 k , RS+3 k , . . . , RS+(n -1) k
- This selection process conceptually arrange population values as follows:

Conceptual framework

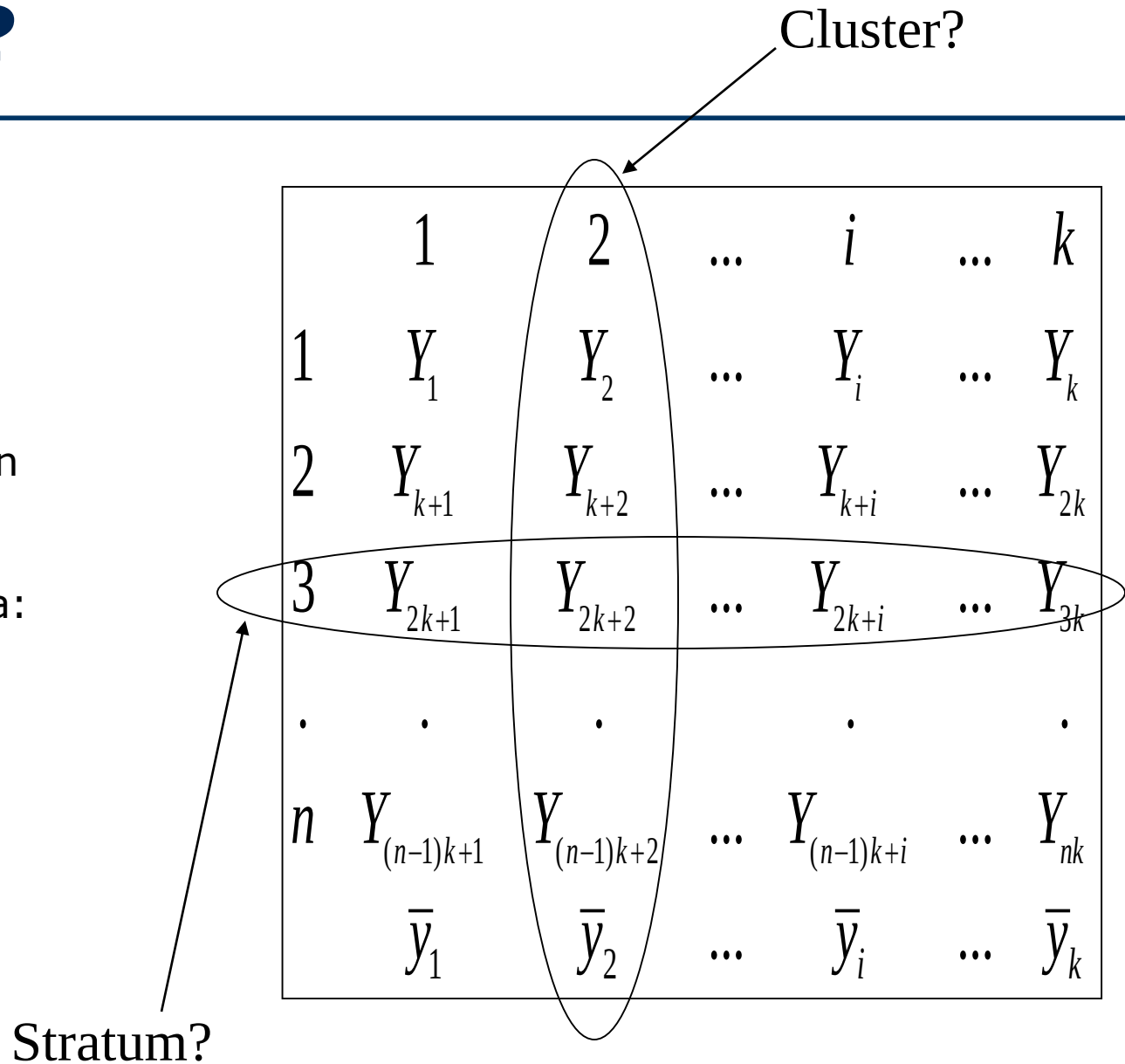


Illustration

	1	2	3	4	5	6
1	1	2	3	4	5	6
2	7	8	9	10	11	12
3	13	14	15	16	17	18
4	19	20	21	22	23	24
5	25	26	27	28	29	30
6	31	32	33	34	35	36
7	37	38	39	40	41	42

Clusters or strata?

- Order of the list achieves stratification
 - Combination of list order and systematic selection
 - Group similar elements together in list
 - Zones in selection similar to strata: contain elements similar to one another
 - Stratified proportionate allocation
- Systematic sampling is not a measurable design:
 - Stratified sample with $n_h=1$
 - Cluster sample with $a = 1$



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B. Fractional intervals

- Interval $k = N/n$ may not be an integer
 - Suppose that $N = 23$ and $n = 5$, so that $k = 23/5 = 4.6$
- If $k = 5$ is applied, $n = 4$ or 5 will be random

	1	2	3	4	5
1	1	2	3	4	5
2	6	7	8	9	10
3	11	12	13	14	15
4	16	17	18	19	20
5	21	22	23		

Solutions for fractional intervals (1)

- What to do?
- Adjust N or n to make k an integer.
- If the achieved sample size is $n+1$, delete a case at random from the sample
 - Unfortunately, this is not *epsem*
- Instead eliminate $N-nk$ listings at random, and then sample systematically
$$f = \left(1 - \frac{N - nk}{N}\right) \times \frac{n}{nk} = \frac{n}{N}$$
- Duplicating $n(k+1)-N$ elements at random also works

Solutions for fractional intervals (2)

- Consider the list to be circular
 - Select a RS from 1 to $N=23$ and continue selection until $n=5$
 - Example: RS=**17**
 - Add integer closest to k ($k=4.6$, so use 5)
 - Select $17+5=\mathbf{22}$
 - $22+5=27$, but $27>23$
 - Go to the beginning of the list and select **4**
 - $4+5=\mathbf{9}$, $9+5=\mathbf{14}$

Solutions for fractional intervals (3)

- Use the fractional interval
 - Choose a random start from 01 to 46 (implied one decimal place)
 - Truncate the decimal to make the selection
 - Add the interval 46 to the random start
 - Truncate the decimal . . . Continue to $n = 5$
 - If the random start is 01-09, the first selection will be obtained after the interval is added
 - Example: RS=35:
 - 35, $35+46=81$, $81+46=127$, $127+46=173$, $173+46=219$
 - Choose 3, 8, 12, 17, 21

Fractional interval summary

RS	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61
	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107
	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153
	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199
	231	232	233	234	235	236	237	238	239						

... ..

RS	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46
	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92
	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138
	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184
	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230

Fractional interval probabilities

Element	Random start	No. of RS's	f
1	10-19	10	1/4.6
2	20-29	10	1/4.6
3	30-39	10	1/4.6
4	40-46,01-03	10	1/4.6
5	04-13	10	1/4.6
...	...	...	...
23	01-09,46	10	1/4.6

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C. Implicit stratification

- List order combined with systematic selection can improve the efficiency (in terms of variance) of systematic sample designs
 - Arrange the list order in advance
 - Determines which samples selected
 - Random order: SRS
 - Stratified order
 - Serpentine order

$$\begin{aligned} n &= 5 \\ k &= 35/5 = 7 \\ RS &= 4 \end{aligned}$$

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D. Variance estimation

- Estimation of the sample mean: $\bar{y} = \frac{\sum y_i}{n}$
- Sampling variance cannot be estimated using only survey data
 - Only a single random start used
 - Two approaches to dealing with the problem
 - Use additional random starts
 - Model the variance
 - Use c random starts and $\bar{y} = \frac{\sum \sum y_{ji}}{cn} = \frac{1}{c} \sum_{\gamma} \bar{y}_{\gamma}$

$$\text{var}(\bar{y}) = \frac{1}{c} \sum_{\gamma} (\bar{y}_{\gamma} - \bar{y})^2$$

Model assisted variance estimation

- Model the population (sample selection process):
- SRS model -- elements in the list are ordered at random.

$$\text{var}(\bar{y}) = (1 - f) \frac{s^2}{n}$$

Stratified random model

- Assumes homogeneity and random ordering within zones
- Proportionately allocated selection with $n_h = 1$ selected per zone
- Collapse neighboring zones to create "pseudo strata" that have more than one selection

$$\text{var}(\bar{y}) = \frac{1-f}{n} \sum W_h s_h^2$$

Paired selection model

- Special case of stratified random model which pairs successive zones
- Simple variance estimation expression
- Simplicity can be disrupted by nonresponse or estimation for a subclass

$$\text{var}(\bar{y}) = \frac{1-f}{n^2} \sum_{h=1}^{n/2} (y_{h1} - y_{h2})^2$$

Successive differences model

- Correlation between the i^{th} and $(i+1)^{\text{th}}$ element larger than the correlation between the i^{th} and $(i+2)^{\text{th}}$ or the i^{th} and $(i+3)^{\text{th}}$, etc.
- $(n-1)$ differences each weighted by $1/2$ (except the first and last)
 - Not unbiased variance estimate, but consistent
 - Bias is small for large n .

$$\text{var}(\bar{y}) = \frac{1-f}{2n(n-1)} \sum_{g=1}^{n-1} (y_g - y_{g+1})^2$$

Variance estimation example

- Take a systematic sample of $n=6$ from the first 60 blocks of the 270 blocks in Kish (p. 625)
 - How many systematic samples of $n=6$ are possible from the first 60 blocks?
 - What is the interval?
- Random start = 3
- What are the 6 selected blocks?

Selected blocks

RS+k	Block	# Rental	# HUs	g
4	242	79	90	1
14	279	84	89	2
24	289	36	45	3
34	378	19	29	4
44	389	21	32	5
54	399	12	18	6

Calculating means

- *Epsem* sample

$$\bar{y}_{\#rental} = \frac{\sum_{i=1}^n y_i}{n}$$

$$= (79 + 84 + 36 + 19 + 21 + 12) / 6 = 41.83$$

Calculating variances (1)

- SRS model

$$\begin{aligned}\text{var}(\bar{y}) &= (1 - f) \frac{s^2}{n} = \left(1 - \frac{6}{60}\right) \left(\frac{1}{6}\right) \frac{(15,539 - 6 * 41.83^2)}{6 - 1} \\ &= (0.90)(0.1667)(1,007.77) = 151.17\end{aligned}$$

Calculating variances (2)

- Paired selection model, even # elements

$$\begin{aligned}\text{var}(\bar{y}) &= \frac{(1-f)}{n^2} \sum_h^{n/2} (y_{ha} - y_{hb})^2 \\ &= \left(1 - \frac{6}{60}\right) \left(\frac{1}{6^2}\right) [(79-84)^2 + (36-19)^2 + (21-12)^2] \\ &= (.9)(0.0278)(25 + 289 + 81) = 9.875\end{aligned}$$

Calculating variances (3)

- Successive differences model

$$\begin{aligned}\text{var}(\bar{y}) &= \frac{1-f}{2n(n-1)} \sum_g^{n-1} (y_g - y_{g+1})^2 \\ &= \frac{1 - \cancel{6}/60}{2 * 6(6-1)} [(79-84)^2 + (84-36)^2 + (36-19)^2 \\ &\quad + (19-21)^2 + (21-12)^2] \\ &= \frac{.9}{60} [25 + 2,304 + 289 + 4 + 81] = 40.55\end{aligned}$$

Choosing among models

- Criteria for selecting an appropriate model include
 - List order
 - random (or "effectively random")
 - ordered deliberately to achieve stratification
 - Degrees of freedom
 - SRS: $n-1$
 - Successive differences: between $n/2$ & $n-1$
 - Use the more conservative $n/2$ in CIs
 - Stratified random: $n-H$
 - Paired differences: $n/2$

Bias considerations

- Variance estimates may be biased
 - Model fails to "capture" stratification represented by systematic selection
 - SRS: no stratification effects are accounted for
 - Stratified selection: captures stratification
 - Collapses across stratum boundaries
 - Paired selection: two per stratum
 - Successive differences: deepest stratification possible